

On-Chain Parametric Cost Insurance for Underserved Latin-American Digital Workers

Record of Closed Iterations and Request for Methodological Guidance

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Purpose

This letter is the entry-point document for the supervisor’s review of the analytics workstream. It records the five iterations that have been closed or parked to date, states the concrete risk target, and asks four practical questions whose answers we expect to bind the next iteration’s pre-registration. A 37-page technical companion at [docs/supervisor-review/2026-05-06-empirical-record-letter](#) provides the per-iteration estimation detail, the verdict-mapping decision tree, and a sample artifact for any question the supervisor wishes to drill into. The implementation venue is on-chain (Panoptic on Uniswap v4) but the empirical work being reviewed here is off-chain econometrics; the on-chain primer and the full methodology are deferred to the companion.

§1. Motivation, problem, suggested solution, challenges

Motivation. We study Latin-American digital workers paid in Colombian pesos who carry recurring USD-denominated professional-input obligations (subscription AI services, API usage fees, cloud compute, software-as-a-service spend), a population for whom the path from wage income into productive-capital ownership has documented structural narrowing ([Mendieta-Muñoz, 2017](#); [Rodrik, 2016](#); [McMillan and Rodrik, 2011](#)).

Problem. When the COP weakens against the USD, the USD-denominated cost line translates into unbudgeted COP expenditure, and no Colombian retail-bank product currently hedges the resulting FX-pass-through cost shock at the hundred-to-low-thousands-USD monthly notional these workers carry.

Suggested solution. We identify the macro risks X that quantitatively produce these cost shocks, validate them empirically, and design parametric cost-insurance products that compensate for the FX-pass-through exposure ([Carter et al., 2017](#); [Mahul and Stutley, 2010](#); [Clarke, 2016](#)). The intermediation product target is a contract that compensates the worker in a unit she actually spends (COP, or a COP-pegged purchasing-parity stablecoin); the implementation venue is on-chain settlement infrastructure (described in the 37-page

technical companion §4), not the economic primitive.

Challenges. • Identification posture — predictive versus causal (see §4.2).

- Convex-payoff sufficiency under insurance-underwriting standard (see §4.2).
- Multi- Y differential calibration; familywise versus false-discovery correction across legs (see §4.2).
- Anti-fishing protocol defensibility — pre-registration thresholds $N_{\text{MIN}} = 75$, $\text{POWER}_{\text{MIN}} = 0.80$, $\text{MDES}_{\text{SD}} = 0.40$ (see §4.2).
- Cross-iteration familywise error and iteration-selection bias (see §4.2).

Bridge to §2. We have run five iterations to date, all closed or parked; the model specifications, results, and failure post-mortems follow in Section 2.

§2. Five iterations

Pair D — BPO offshoring through COP/USD lag (PASS). Colombian young workers (14–28) employed in the broad CIU Rev. 4 services aggregate (Sections G–T), with Y_t the logit of the monthly young-worker share on DANE GEIH micro-data and X_{t-k} the lagged log Banrep TRM end-of-month spot.

$$Y_t = \alpha + \beta_6 X_{t-6} + \beta_9 X_{t-9} + \beta_{12} X_{t-12} + \varepsilon_t$$

where $Y_t = \text{logit}(\text{share}_t^{\text{GT}})$ and $X_{t-k} = \log(\text{COP/USD}_{t-k})$ for $k \in \{6, 9, 12\}$ months; the composite is $\hat{\beta}_{\text{composite}} = \hat{\beta}_6 + \hat{\beta}_9 + \hat{\beta}_{12}$. Data: `notebooks/bpo_offshoring_fx_lag/` (governing spec sha `964c62cca0...ef659`, $N = 134$ monthly, 2015-01-31 through 2026-02-28). Result: $\hat{\beta}_{\text{composite}} = +0.13670985$, $\text{HAC}(L = 12)$ standard error 0.02465, $t = +5.5456$, one-sided $p \approx 1.46 \times 10^{-8}$; robustness rows R1–R4 produced zero sign flips and the four-row consistency classifier returned AGREE. *Interpretation.* The lagged COP/USD log-level positively predicts the broad-services young-worker share at the pre-registered horizons, clearing the gate. *Caveats inherited from independent review.* The estimate hedges a correlation, not a causal identification of the BPO transmission channel; the lag effect is concentrated at the six-month horizon ($\hat{\beta}_6 \approx +0.109$, about 80% of the composite) rather than spread across the 6–12-month window; the post-2014 window over-represents the oil-collapse, COVID, and 2022–2024 Fed-tightening regimes; the verdict is sensitive to a single off-spec design choice (an orchestrator-brief variant including a `marco2018_dummy` in the primary returns $\hat{\beta}_{\text{composite}} = +0.0815$ at $p \approx 0.080$); and the dev-AI Stage-1 R2 Section-M reframe (below) suggests the operative sub-aggregate is consultancy/scientific/administrative services rather than Section-J ICT-narrow tech offshoring.

dev-AI Stage-1 — Section J ICT through COP/USD lag (FAIL). Colombian young workers (14–28) restricted to CIU Rev. 4 Section J (Information and Communication, Divisions 58–63), a strict subset of the Pair D cut, with the same monthly cadence, `FEX_C_2018` expansion

factors, and lag panel.

$$Y_t = \alpha + \beta_6 X_{t-6} + \beta_9 X_{t-9} + \beta_{12} X_{t-12} + \varepsilon_t$$

where $Y_t = \text{logit}(\text{share}_t^J)$ and $X_{t-k} = \log(\text{COP}/\text{USD}_{t-k})$; pre-registered $H_1 : \beta_{\text{composite}} > 0$. Data: `notebooks/dev_ai_cost/` (governing spec sha `7c72292516...751f5a`, $N = 134$ monthly, 2015-01-31 through 2026-03-31). Result: $\hat{\beta}_{\text{composite}} = -0.14613$, $\text{HAC}(L = 12)$ standard error 0.0847, $t = -1.726$, one-sided $p \approx 0.958$ against $H_1 : \beta > 0$; R-consistency classifier MIXED with three of four R-rows agreeing on the negative primary sign. *Interpretation.* The pre-registered positive-sign expectation is rejected at the Section-J narrow cut; this is a sign-flipped FAIL, not an underpowered failure-to-reject. *Why it fails.* The Section-J substrate does not exhibit the lagged-FX-pass-through-to-employment pattern the BPO-offshoring narrative pre-supposed; an empalme-residual bias surfaced (the Phase-1 boundary diagnostic returned a $+0.375$ logit- Y jump at the 2020-12 $\rightarrow$ 2021-01 Marco-2018 boundary against a ± 0.335 envelope), so the post-2021 level shift is not fully neutralized at the Section-J cut. The R2 Section-M sensitivity arm (CIU Sections 69–75) returned $\hat{\beta}_{\text{composite}} = +0.45482801$, $t = +4.73$, $p = 1.13 \times 10^{-6}$, but this is a not-pre-registered candidate-next-iteration object, not a graduated PASS, and the rejection at Section-J falsifies the per-user instrument design that the dev-AI-paying population would have hosted.

FX-vol-CPI surprise — Colombian CPI through TRM realised volatility (FAIL).

Colombian corporates and remittance recipients with FX-volatility exposure on COP/USD around DANE CPI publication, weekly cadence on a Friday anchor, with Y_w the weekly realised volatility of the Banrep TRM daily log-return panel and the regressor the lagged signed AR(1) CPI surprise.

$$\text{RV}_w^{\text{COP/USD}} = \alpha + \beta \text{Surp}_{w-1}^{\text{CPI}} + \varepsilon_w$$

where RV_w is realised volatility in week w and $\text{Surp}_{w-1}^{\text{CPI}}$ is the lagged signed monthly CPI surprise mapped to weekly cadence (carried across the release week, zero otherwise), with a small set of macro-surprise controls in the primary. Data: `notebooks/fx_vol_cpi_surprise/` (panel fingerprint `nb1_panel_fingerprint.json`, $N = 947$ weekly, 2008-01-02 through 2026-02-23). Result: $\hat{\beta}_{\text{CPI}} = -0.000685$, $\text{HAC}(4)$ standard error 0.001794, 90% HAC interval $[-0.003636, +0.002266]$ contains zero; the gate criterion $\hat{\beta} - 1.28 \cdot \widehat{\text{SE}} = -0.002981 < 0$ fires the FAIL. *Interpretation.* Under the pre-committed weekly OLS specification, signed AR(1) CPI surprises do not produce a statistically detectable increase in COP/USD realised volatility across the sample window. *Why it fails and sensitivity caveat.* The pre-registered T1 exogeneity test rejects at $F = 15.12$, $p \approx 1.3 \times 10^{-9}$, so the iteration is a predictive regression rather than a structural-causal one and $\hat{\beta}$ cannot be read as an impulse-response. Two pre-registered sensitivities returned positive significant 90% intervals at the one-sided $\alpha = 0.05$ convention — A1 monthly cadence $\hat{\beta} = +0.0152$ on $[+0.0057, +0.0246]$; A4 release-day-excluded $\hat{\beta} = +0.0033$ on $[+0.0005, +0.0062]$ — but neither is the iteration's main finding: the pre-registered weekly primary is the result, and A1/A4 sit as candidate inputs to a possible future monthly-cadence-primary specification rather than as a rescue of the weekly primary.

Phase-A.0 Remittance — Colombian remittance corridor through FX (EXIT_NON_REMITTANCE)

Colombian household remittance recipients indexed against the Colombian remittance corridor with FX pass-through, with Y_w inherited from the FX-vol weekly panel and X_{w-1} a candidate on-chain remittance-flow proxy constructed from cCOP and COPm aggregate user activity on Celo.

$$RV_w^{\text{COP/USD}} = \alpha + \beta \Delta \text{Remit}_{w-1}^{\text{Col}} + \varepsilon_w$$

where $\Delta \text{Remit}_{w-1}^{\text{Col}}$ is the lagged Colombian household-remittance flow change as proxied on Celo on-chain activity, partitioned to attempt to isolate household remittance from third-party-DEX activity, treasury roundtripping, and bot arbitrage. Data: Phase-A.0 Rev 4.1 spec chain (kill-criterion-bearing); window 2024-09 through 2026-04, weekly cadence (panel construction did not reach final N before the EXIT verdict fired). Result: no empirical estimate produced — exited on kill criteria pre-estimation. *Interpretation.* The pre-registered remittance specification was *never estimated*: kill criterion k1 (intent: an X-driver fingerprint must be present in remittance event-window data) and partial k2 (data-source: the X proxy must measure Colombian household remittance, not a generic on-chain volume aggregate) both fired before estimation, producing the EXIT_NON_REMITTANCE label on 2026-04-24. *Why it exits.* Task 11.F Axis-1 event research returned zero of thirty peak-day flows fingerprinting as remittance, with approximately 87% of surveyed flow decomposing into TRM arbitrage, treasury roundtripping, campaigns, and bot activity; the cCOP/COPM aggregate user activity on Celo is third-party-DEX volume, not Colombian household remittance, so no post-hoc filter could reconstruct an X-driver that was never measured upstream. The exit is load-bearing on the kill-criterion discipline: absent the pre-registered EXIT, the rev-4 plan's non-stop filter iteration would have run to a filter that appeared to “rescue” a signal that was never captured.

P1 Bittensor SN18 — event-study apparatus (PARKED). The proprietary-AI policy-shock-exposed population (LATAM developers paying USD-denominated AI APIs as the downstream candidate; the apparatus pre-dates target-population narrowing), with Y_t a Bittensor SN18 (Cortex.t) alpha event-study indicator and X_t a set of pre-pinned policy-event dates (Bittensor halving 2025-12-15; dTAO mainnet activation 2025-02-13; Cortex.t milestones).

$$\text{AR}_{i,\tau} = \alpha_i + \sum_{\tau=-L}^{+L} \gamma_\tau \mathbf{1}\{t_i + \tau \in \mathcal{E}\} + \varepsilon_{i,\tau}$$

where $\text{AR}_{i,\tau}$ is the event-window abnormal alpha indicator at relative time τ for event i , $\mathcal{E}$ is the pre-pinned policy-event set, and the apparatus is evaluated over a nine-cell verdict cube (verdict-eligibility crossed with robustness-consistency), Bonferroni-corrected at $\alpha_{\text{primary}} = 0.0167$ over the conjunctive family P1+P2+P3 with $N_{\text{min}}^{\text{events}} = 8$ and an asymmetric placebo gate. Data: spec sha `f855e036d3...b47aeab`; apparatus completed, data ingest deferred. Result: none — apparatus parked for the record on 2026-04-27; no graduating empirical claim produced. *Interpretation.* The slow-lane apparatus (three spec revisions, six verifier dispatches, a nine-cell verdict cube) was overkill relative to the project's stage goal of a single empirically validated risk before protocol design. *Why parked.* Apparatus complete; no graduating empirical claim. The user pivoted to the simple-composite- β fast-lane track (Pair D, dev-AI Stage-1) and parked P1

to preserve the intellectual capital of the spec work, with re-activation conditions documented in the parking memory.

§3. Risk-target framing

The risk target is FX-volatility exposure for a specific population: Latin-American digital workers paid in Colombian pesos (COP cash-flows) but carrying recurring USD-denominated professional-input obligations. When the COP weakens against the USD, the USD-denominated cost line translates into an unbudgeted COP-expenditure increase. The intermediation product target is a contract that compensates for this FX-pass-through cost shock, denominated in a unit the worker actually spends (COP, or a COP-pegged purchasing-parity stablecoin). The Abrigo Lemma's abstract $Y_{\text{inequality}}(t) = R_a(t) - R_c(t)$ framing — present in the 37-page technical companion §3 — is the eventual analytical generalisation; the entry-point letter introduces the concrete target only.

§4. Asks

§4.1 Suggested specifications

We invite the supervisor to suggest the (Y, X) pair, lag structure, sample window, primary inference convention, and robustness arms for the next iteration. Specifically: which sectoral cut should follow the Pair D / dev-AI G–T / J split given the R2 Section M signal; whether to repeat the FX-vol-on-CPI-surprise primary at A1 monthly cadence as the next pre-registration; or whether to pivot to a non-Colombian LATAM panel.

§4.2 Risks

The bullets below are risks the project has already identified; we ask the supervisor to add risks we have not flagged.

- Identification posture — predictive-regression versus structural-causal; T1 exogeneity rejected on FX-vol; HAC fixed- L per iteration.
- Convex-payoff sufficiency under insurance-underwriting standard — mean- $\hat{\beta}$ is necessary but not sufficient; [Carter et al. \(2017\)](#), [Mahul and Stutley \(2010\)](#), and [Clarke \(2016\)](#) set the standards.
- Multi- Y differential calibration — joint-leg correlation, FWER versus FDR, regime-switching.
- Anti-fishing protocol defensibility — $N_{\text{MIN}} = 75$, $\text{POWER}_{\text{MIN}} = 0.80$, $\text{MDES}_{\text{SD}} = 0.40$ inherited from Phase-A.0 Rev-5.3.1, not re-derived.
- Cross-iteration FWER and iteration-selection bias — five closed iterations; meta-multiple-comparison concern; [Heckman \(1979\)](#) as candidate frame.
- Stambaugh small-sample bias on persistent-regressor predictive regressions — [Stambaugh \(1999\)](#), [Phillips \(2014\)](#); log-FX is near-unit-root at monthly frequency.

- Empalme residual bias — DANE Marco-2018 methodology break not fully neutralised; surfaced in dev-AI Stage-1 NB02 Trio 1 boundary anomaly.

§4.3 Bibliography

The current bibliography is documented in the 37-page companion's Appendix B and is reproduced as `refs.bib`. We invite the supervisor to recommend additional references he considers load-bearing for the framework's empirical work, particularly on (a) parametric-index insurance pricing in low-counterparty-credit-standing populations, (b) sectoral employment-share predictive regressions on real-effective-exchange-rate panels, and (c) cross-iteration false-discovery-rate correction regimes for observational pre-registration.

§4.4 Connections

We invite the supervisor to introduce the project to other researchers, practitioners, or institutions whose work might complement this iteration roster — including but not limited to parametric-insurance underwriters with LATAM exposure, FX-derivatives operators serving sub-corporate notionals, on-chain settlement engineers, and applied econometricians working on premature deindustrialization in Latin America. The framework's empirical work is open to collaboration; the supervisor's network is the most efficient channel for identifying complementary contributors.

Closing and pointer to the companion

Five iterations are closed or parked; no Stage-1 work is in progress. The next iteration's pre-registration is held open pending the supervisor's response to the four asks above. The 37-page technical companion at `docs/supervisor-review/2026-05-06-empirical-record-letter.pdf` contains the full per-iteration record (companion §6, nine-field schema per iteration), the on-chain settlement primer (§4), the methodology in full (§5, including the verdict-mapping decision tree), and four appendices: Glossary (A), Cited Literature (B), Pair D Verdict Header sample artifact (C), and Power-Calculation Table (D). The supervisor is invited to drill into any iteration's per-iteration block in companion §6 for the specification chain, the sha256 pre-registration governance, and the independent-review caveats inherited from the closure memoranda.

References

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